# OspSimulator Software Requirements Specification

For Software Version 1.0.0

# Table of Contents

# 1 Introduction

This document is the software requirements specification for an application called the OspSimulator. This application is a test program that allows the user to generate Next Generation 9-1-1 (NG9-1-1) calls that simulate emergency calls as they should be provided by an NG9-1-1 Originating Service Provider (OSP).

## 1.1 References

This section lists the standards to be used for the development of this application by category. The fact that a standard is listed as a reference here does not imply that all requirements in that standard will be implemented. Sections 2 through 5 present the specific requirements from these standards that shall be implemented.

### 1.1.1 NENA Standards

1. NENA i3 Standard for Next Generation 9 1 1, NENA, NENA STA-010.3f-2021, October 7, 2021.

### 1.1.2 General SIP RFCs

2. SIP: Session Initiation Protocol, IETF, RFC 3261, June 2002.
3. An Offer/Answer Model with the Session Description Protocol (SDP), IETF, RFC 3264, June 2002.
4. A Presence Event Package for the Session Initiation Protocol (SIP), IETF, RFC 3856, August 2004.
5. SDP : Session Description Protocol, IETF, RFC 4566, July 2006.
6. RTP: A Transport Protocol for Real-Time Applications, IETF, RFC 3550, July 2003.
7. RTP Payload for DTMF Digits, Telephony Tones, and Telephony Signals, IETF, RFC 4733, December 2006.

### 1.1.3 NG9-1-1 Specific RFCs/Standards

8. Additional Data Related to an Emergency Call, IETF, RFC 7852, July 2016.
9. Framework for Emergency Calling Using Internet Multimedia, IETF, RFC 6443, December 2011.
10. Best Current Practice for Communications Services in Support of Emergency Calling, IETF, RFC 6881, March 2013.
11. Location Conveyance for the Session Initiation Protocol, IETF, RFC 6442, December 2011.
12. Next-Generation Pan-European eCall, IETF, RFC 8147, May 2017.
13. Next-Generation Vehicle-Initiated Emergency Calls, IETF, RFC 8148, May 2017.
14. Advanced Automatic Collision Notification (AACN) Vehicle Emergency Data Set (VEDS), NENA/APCO, APCO/NENA Candidate ANS 2.102.1.2022.
15. Common Alerting Protocol Version 1.2, OASIS, July 2010.
16. HTTP-Enabled Location Delivery (HELD), IETF, RFC 5985, September 2010.
17. Use of Device Identity in HTTP-Enabled Location Delivery (HELD), IETF, RFC 6155, March 2011.

### 1.1.4 Security Related RFCs

18. The Secure Real-time Transport Protocol (SRTP), IETF, RFC 3711, March 2004.
19. Session Description Protocol (SDP) Security Descriptions for Media Streams, IETF, RFC 4568, July 2006.
20. The Use of AES-192 and AES-256 in Secure RTP, IETF, RFC 6188, March 2011.

21. Framework for Establishing a Secure Real-time Transport Protocol (SRTP) Security Context Using Datagram Transport Layer Security (DTLS), IETF, RFC 5763, May 2010.
22. Datagram Transport Layer Security (DTLS) Extension to Establish Keys for the Secure Real-time Transport Protocol (SRTP), IETF, RFC 5764, May 2010.

### 1.1.5 MSRP RFCs

23. The Message Session Relay Protocol (MSRP), IETF, RFC 4975, September 2007.
24. Common Profile for Instant Messaging (CPIM), IETF, RFC 3860, August 2004.
25. TCP-Based Media Transport in the Session Description Protocol (SDP), IETF, RFC 4145, September 2005.
26. An Alternative Connection Model for the Message Session Relay Protocol (MSRP), IETF, RFC 6135, February 2011.

### 1.1.6 RTT RFCs

27. RTP Payload for Text Conversation, IETF, RFC 4103, June 2005.

# 2 General Requirements

This application shall be capable of delivering a single call at a time to a functional element under test.

Outgoing calls shall delivered over an IP network using the SIP protocol.

## 2.1 Basic Functional Requirements

### 2.1.1 User Call Controls

The application shall provide the following call control functions.

1. Initiate a new call
2. Terminate a call
3. Cancel a call request before it is answered

### 2.1.2 Multi-Media Call Support

The application shall support the following media types.

1. Audio
2. Video
3. Real Time Text (RTT)
4. Message Session Relay Protocol (MSRP)

The user shall have the capability of selecting what media types to offer when it initiates a call.

The user shall have the ability to add a new media type to an established call.

The application shall be capable of accepting a re-INVITE request from the called party to add a new media type to an established call.

#### *2.1.2.1 Audio Codec Support*

The application shall provide support for the following audio codecs.

1. G.711 Mu-Law
2. G.711 A-Law
3. G.722
4. G.729
5. AMR-WB

The user shall be able to select which audio codecs to offer for an outgoing call.

The user shall be able to specify the order that the audio codecs are offered.

#### *2.1.2.2 Video Codec Support*

The application shall support the following video codecs.

1. H.264
2. VP8

The user shall be able to select which video codecs to offer for an outgoing call.

The user shall be able to specify the order that the video codecs are offered.

## 2.2 Location and Additional Data Support

The application shall be capable of generating call requests that provide call location and NG9-1-1 additional data. The application shall allow the user to create a database of information that can be sent with a call. The database shall allow the user to associate location data and NG9-1-1 additional data with the caller ID of a call.

The application shall provide a default database of location and additional data. The default database shall provide a general framework so that users can add or modify it.

Each individual Windows user shall be able to modify the default database or add new call related data to it. A user's database shall be located in the application data directory for the user. For example, it the user's Windows user name is John, the additional data and location database shall be:

C:\Users\John\AppData\Local\OspSimulator\AdditionalData.

If the AdditionalData directory for a current Windows user does not exist when the OspSimulator application starts up, the application shall copy the default database to the AdditionalData directory for that user. The user can then modify the data under the user specific AdditionalData directory.

The AdditionalData directory shall contain subdirectories. Each subdirectory shall be a 10-digit telephone number that identifies the simulated call's calling party number. Each subdirectory shall contain one or more XML files that contain location or additional data.

The following subsections specify the capabilities that shall be available for location and additional data.

Section 4 specifies the requirements for the location service (HELD server) that the application must support. Section 5 specifies the requirements for the additional data service that the application must support.

### 2.2.1 Location Data

Call location data shall be stored as XML files using the Presence Information Data Format with Location Object (PIDF-LO) schema.

The user shall have the option of specifying the following methods of sending PIDF-LO data with a call.

1. Location by-value
2. HTTP Enabled Location Data (HELD, RFC 5985)
3. A Presence Event Package for the Session Initiation Protocol (SIP) (RFC 3856)

The user shall be able to specify all three methods of location delivery.

The user shall be able to disable all location delivery for outgoing calls.

For location by-reference (HELD), the user shall be able to specify the following.

1. Transport Type: HTTP or HTTPS
2. IP Network Type: IPv4 or IPv6

### 2.2.2 Additional Data

The application shall support the following types of NG9-1-1 additional data.

1. Provider Information (see Section 4.1 of RFC 7852)
2. Service Information (see Section 4.2 of RFC 7852)
3. Device Information (see Section 4.3 of RFC 7852)
4. Subscriber Information (see Section 4.4 of RFC 7852)
5. Comment (see Section 4.5 of RFC 7852)
6. Advanced Automatic Crash Notification (see RFC 8148)

The application shall support both location by-value and location by-reference. The user shall be able to specify the delivery method as follows.

1. No additional data
2. By-value
3. By-reference
4. By-Value and By-Reference

For additional data by-reference, the user shall be able to specify the following.

1. Transport Type: HTTP or HTTPS
2. IP Network Type: IPv4 or IPv6

## 2.3 Network Configurations

The application shall be capable of operating in the following network environments.

1. IPv4 and IPv6
2. IPv4 only
3. IPv6 only

The network configuration shall be configurable by the user.

For each network type, the application shall listen on a single selected IP address.

## 2.4 Call Addressing Requirements

The user shall be able to specify a fully qualified SIP URI that will be used to determine how to route outgoing call requests. The SIP URI may be the final call endpoint or an intermediate proxy such as a BCF or an ESRP.

The host portion of the SIP URI may be either an IP address (IPv4 or IPv6) or the host's domain name. If the host portion of the SIP URI is not an IP address then the application shall perform an A record or an AAAA record DNS lookup to determine the IP address of the host.

## 2.5 SIP Transport Support

The application shall support the following SIP transport protocols.

1. UDP
2. TCP
3. SIPS (SIP over TLS)

The software shall automatically determine the SIP transport protocol to use based on the To SIP URI described in the previous section.

## 2.6 Operational Environment

### 2.6.1 Windows Operating System

The application shall be designed to work on Windows 10 Professional or later.

The application is not required to run on the Home editions of the Windows operating systems but it may run on a Windows Home Edition PC with “S” (secure) mode turned off.

### 2.6.2 Minimum PC Requirements

The application shall run on a Windows PC that provides the following.

1. Display: 1920x1080 pixels minimum
2. Audio: Speakers and microphone or headset
3. Camera: Built-in or external via USB. Up to VGA resolution (640x480 pixels)
4. Memory: 8 GB minimum
5. Pointing Device: Mouse or touch-pad.

The application may run on a Windows PC that has a 4K display but the user interface does not need to be optimized to support such a high screen resolution.

The application may run on a Windows tablet PC with a touch screen but the user interface design of the application does not need to be optimized for touch screen operation.

### 2.6.3 Application Installation Requirements

The installation technology shall be a Windows setup MSI package. The setup program shall automatically install any prerequisites (such as .NET).

A self-extracting EXE installation program shall be developed.

The setup program shall configure Windows to run this application as a Windows administrator.

The installation program shall install a shortcut icon on the user’s desktop.

The installation program and the application itself does not need to be signed with a code signing X.509 certificate.

# 3 NG9-1-1 SIP Call Interface Requirements

## 3.1 SIP Method Support

The following table specifies which Session Initiation Protocol (SIP) methods the application must support.

| SIP Method | NENA-STA-010.3f Requirement | Supported? | Comments |
|---|---|---|---|
| INVITE | Mandatory | Yes | |

| | | | |
|---|---|---|---|
| REFER | Mandatory | No | The application does not need to be able to setup a conference or add additional parties to the call. |
| BYE | Mandatory | Yes | |
| CANCEL | Mandatory | Yes | |
| UPDATE | Mandatory | No | Used for sending updated SIPREC metadata to SIPREC recorders only. Not used for incoming calls.<br><br>This application does not need to support SIPREC. |
| OPTIONS | Mandatory | No | |
| ACK | Mandatory | Yes | |
| PRACK | Mandatory | No | Future |
| MESSAGE | Mandatory | No | Used for incoming non-interactive Common Alerting Protocol (CAP) calls only.<br><br>This application does not need to support non-interactive calls. |
| INFO | Mandatory (but must not be used to convey DTMF digits) | No | |
| REGISTER | Must not support | No | |
| SUBSCRIBE | Mandatory | Yes | The application shall accept SUBSCRIBE requests for the Presence Event Package for the Session Initiation Protocol (SIP) (RFC 3856). |
| NOTIFY | Mandatory | Yes | Used for RFC 3856 support. |
| PUBLISH | Optional | No | Not used for NG9-1-1 applications |

## 3.2 Emergency Call-ID and Emergency Incident-ID

The software shall add a Call-Info header containing an emergency call identifier as specified in Section 2.1.6 of NENA-STA-010.3f to the outgoing INVITE request.

The software shall add a Call-Info header containing an emergency incident identifier as specified in Section 2.1.7 of NENA-STA-010.3f.

## 3.3 Support for Re-INVITE Requests

The application shall be able to handle incoming Re-INVITE requests for the currently on-line call.

Re-INVITE requests may be sent by the called party to perform the following functions.

1. Change the destination endpoint (IP address and media port number)
2. Change the characteristics of the media transport for one or more media types, such as the type of media encryption.

3. Add new media to the call.

## 3.4 Sending Re-INVITE Requests

The application shall be capable of sending re-INVITE requests to the called party for the purpose of adding new media to the current call.

## 3.5 Sending Offer-less INVITE Requests

The application shall provide a configuration setting that allows the user to send a call with an offer-less INVITE request. In an offer-less INVITE call, the application shall send an initial INVITE request that does not contain a SDP block in the body of the request. When the remote endpoint sends an OK response containing an SDP block in the response body, the application sends its answer SDP in the ACK request that it sends in response to the OK response that it received.

## 3.6 Calls with No Media

The user shall have the ability to enable or disable each type of media for the initial INVITE request of an outgoing call in the application configuration settings. If the user starts a new call when no media is enabled, the application shall send an INVITE request with an SDP offer that contains no media blocks. The user or the remote endpoint shall then be able to add media to the call using the re-INVITE mechanisms described above.

The expected use case of a call without any media is an Advanced Automatic Crash Notification (AACN) call that is generated by a vehicle's automatic crash detection system. Such calls may have an AACN additional data information block along with location information attached to the initial call request.

# 4 Location Information Service Requirements

The OspSimulator application shall implement the server side of the HELD protocol as specified in RFC 5985.

If location by-reference is enabled, the application shall construct a Geolocation SIP header and add it to the outgoing INVITE request. The Geolocation SIP header shall contain an HTTP or HTTPS URI that will allow the called party to retrieve location data using the HELD protocol.

The general pattern of the request string for location data shall be:

    http://IPEndPoint/Location/CallingPartyNumber

The application shall listen on either HTTP or HTTPS depending on the "UseHttps" setting described in Section 6.

The "IPEndPoint" portion of the request string specifies the host. The address portion of the IPEndPoint shall be either an IPv4 or an IPv6 address depending upon the UseIPv4ForHttp setting described in Section 6.

The "CallingPartyNumber" request path parameter shall specify the 10-digit telephone number of the selected calling party for a call. This path parameter specifies the name of a directory under the AdditionalData directory that contains the location data which is stored as a PIDF-LO XML document.

The application shall support the POST method for HELD requests.

The application shall also support the GET method for HELD requests. This requirement is for testing purposes only.

# 5 Additional Data Service Requirements

The OspSimulator application shall implement the server side of the additional data protocol as specified in RFC 7852.

If additional data by-reference is enabled then the application shall build and attach SIP Call-Info headers to the outgoing INVITE request for each type of additional data that is available for the call. Each Call-Info header shall contain an HTTP or HTTPS URI that will allow the called party to de-reference it to obtain the additional data.

The application shall support the GET method for de-referencing additional data.

The general pattern for the request string for additional data shall be:

```
http://IPEndPoint/AdditionalData/CallingPartyNumber/InfoType
```

The application shall listen on either HTTP or HTTPS depending on the "UseHttps" setting described in Section 6.

The "CallingPartyNumber" request path parameter shall specify the 10-digit telephone number of the selected calling party for a call. This path parameter specifies the name of a directory under the AdditionalData directory that contains the additional data which is stored as an XML document.

The "InfoType" path parameter specifies the type of additional data being requested as specified in the following table.

| InfoType Path Parameter | Description |
|---|---|
| Comments/CommentID | CommentID specifies the name of a comments XML file located under the call's additional data directory. For example: Comments1.xml. This allows the application to provide multiple comments for a call. |
| DeviceInfo | Specifies the DeviceInfo.xml file for the call. |
| Providers/ProviderID | ProviderID specifies the name of the provider information XML file located under the call's additional data directory. For example: Provider1.xml. This allows the application to provide multiple provider information additional data blocks of information for a call. |
| ServiceInfo | Specifies the ServiceInfo.xml file for the call. |
| SubscriberInfo | Specifies the SubscriberInfo.xml file for the call. |
| AutomatedCrashNotification | Specifies the AutomatedCrashNotification.xml file for the call. |

# 6 Application Configuration Settings

Application configuration settings shall be stored as JSON files underneath the Windows special folder called MyDocuments. For example, if the current user is John, the configuration files shall be stored in C:\Users\John\OneDrive\Documents\OspSimulator.

## 6.1 Network Settings

| Setting | Value Type | Default | Description |
|---|---|---|---|
| EnableIPv4 | Boolean | True | If true, then the application will use IPv4. |
| EnableIPv6 | Boolean | True | If true, then the application will use IPv6. |
| UseMutualAuthentication | Boolean | True | Applies to SIP over TLS. If true, then the application shall offer its X.509 certificate when it attempts to connect to the remote host using SIP over TLS. |
| UseHttps | Boolean | True | If true, then the application shall listen on HTTPS for its HELD and additional data interfaces. If false, then the application will listen on HTTP. |
| UseIPv4ForHttp | Boolean | True | If true, then use IPv4 for the HTTP interfaces, else use IPv6 for the HTTP interfaces. |
| HttpPortNumber | Numeric | 11000 | Specifies the TCP port number to use for HTTP interfaces. |
| HttpsPortNumber | Numeric | 11001 | Specifies the TCP port number to use for HTTPS interfaces. |
| LocalSipPortNumber | Numeric | 5060 | SIP port for UDP and TCP. |
| LocalSipsPortNumber | Numeric | 5061 | SIP port for Transport Layer Security (TLS). |
| IPv4Address | String | NA | The default will be the last selected IP address for the IPv4 network if it's available, or the first IP address in the list of available IP addresses for the IPv4 network. |
| IPv6Address | String | NA | The default will be the last selected IP address for the IPv6 network if it's available, or the first IP address in the list of available IP addresses for the IPv6 network. |
| MediaPorts | NA | NA | See Media Port Ranges below. |

### 6.1.1 Media Port Ranges

The application shall provide the following media port settings for each media type.

1. Starting Port
2. Number of Ports

The following table specifies the default media port settings.

| Media Type | Starting Port | Number of Ports |
|---|---|---|
| Audio | 6000 | 1 |
| Video | 7000 | 1 |
| RTT | 8000 | 1 |
| MSRP | 9000 | 1 |

The default for the Number of Ports setting is 1 because the application only supports one call at a time.

## 6.2 Certificate Settings

The application requires an X.509 certificate so that it can act as a client for SIP TLS and server HTTPS requests.

The application shall provide a default X.509 PFX file that will be installed by the installation program.

The user shall be allowed to provide a custom X.509 PFX file by specifying the following.

1. X.509 Certificate Path
2. X.509 Certificate Password

The user shall be able to restore the certificate to the default X.509 certificate file.

## 6.3 Call Handling Settings

### 6.3.1 Last Call Address Settings

| Setting | Value Type | Default | Description |
|---|---|---|---|
| SipToUri | String | Null | Contains the last To SIP URI entered by the user. The To SIP URI is used to call the remote party. |
| FromUser | String | Null | Contains the last From Number entered by the user. The From Number is the user part of the local SIP URI in the From header of the outgoing INVITE request. |
| UseUrnServiceSos | Boolean | False | If true, then the request URI of the outgoing INVITE request shall be set to “urn:service:sos”. If false then the request URI of the outgoing INVITE request will be set to the To SIP URI. |
| UseTelUri | Boolean | False | If true, then the application uses a “tel” URI in the SIP From header of the outgoing call. |
| PreferIPv6 | Boolean | True | This setting is applied if the application needs to do a DNS host name lookup for the remote host and the DNS server returns an IPv6 address for the host. |

## 6.4 Media Settings

### 6.4.1 Enabled Media Settings

The Enabled Media Settings specify which media types the application will offer in an outgoing INVITE request.

| Setting | Value Type | Default | Description |
|---|---|---|---|
| OfferAudio | Boolean | True | If true then the application will offer audio media with the initial INVITE request. |
| OfferVideo | Boolean | False | If true then the application will offer video media with the initial INVITE request. |
| OfferRtt | Boolean | False | If true then the application will offer Real Time Text (RTT) with the initial INVITE request. |
| OfferMsrp | Boolean | False | If true then the application will offer Message Session Relay Protocol (MSRP) with the initial INVITE request. |

### 6.4.2 Media Codec Settings

The media codec settings specify which audio and video codecs the application will offer in an outgoing INVITE request. The user shall be able to customize each list by adding or removing codecs. The user shall be able to determine the order of the codecs in each list. Each list must contain at least one codec.

| Setting | Value Type | Default | Description |
|---|---|---|---|
| AudioCodecs | List<string> | PCMU, PCMA, G722, G729, AMR-WB | Contains a list of audio codecs that the application will offer for audio in the outgoing INVITE request. |
| VideoCodecs | List<string> | H264, VP8 | Contains a list of video codecs that the application will offer for video in the outgoing INVITE request. |

### 6.4.3 Media Encryption Settings

These settings specify the type of media encryption to offer for outgoing calls.

The following options shall be available for RTP type media (audio, video and RTT).

1. None
2. SDES-SRTP
3. DTLS-SRTP

The default setting shall be "None".

The following options shall be available for MSRP media.

1. None
2. MSRPS (MSRP over TLS)

The default setting shall be "None".

### 6.4.4 Miscellaneous Media Settings

| Setting | Value Type | Default | Description |
|---|---|---|---|
| UseMsrpCpim | Bool | false | If true then the application will always use messabe/cpim for sending MSRP text messages. |
| UseRecordedAudio | Bool | false | If true, then the application will send audio from an audio recording file instead of audio captured from the microphone. If false, then the application shall send audio captured from the microphone. |
| UseDefaultAudioRecording | Bool | true | If true then the application shall use audio from a default audio recording file. Applies only if UseRecordedAudio is true. |
| AudioRecordingFilePath | String | Empty | Specifies the full path for a user-provided audio recording file. Used only if UseRecordedAudio is true and UseDefaultAudioRecording is false. |

## 6.5 Location and Additional Delivery Settings

### 6.5.1 Location Delivery Settings

These settings specify how the application will deliver call location to the called party. The settings are independent. The user may select any combination of methods. If no location delivery method is selected then the application will not deliver location data for the outgoing call.

| Setting | Value Type | Default | Description |
|---|---|---|---|
| LocationByValue | Boolean | True | If true, then location data will be delivered in the body of the SIP INVITE request that is sent as specified in Section 3.1 of RFC 6442. |
| LocationByReference | Boolean | False | If true, then location data is delivered by-reference as specified in Section 3.2 of RFC 6442. |
| LocationByPresenceEvent | Boolean | False | If true, then location data will be delivered by the SIP Presence Event Package as specified by Section 3.3 of RFC 6442 and RFC 3856. |

### 6.5.2 Additional Data Delivery Settings

These settings specify how the application will deliver additional call data to the called party. The settings are independent. The user may select any combination of delivery methods. If no delivery method is selected then the application will not deliver additional data for the outgoing call.

| Setting | Value Type | Default | Description |
|---|---|---|---|

| | | | |
|---|---|---|---|
| AdditionalDataByValue | Boolean | False | If true, then additional data will be delivered by-value in the body of the SIP INVITE request that is sent. See Section 6 of RFC 7852. |
| AdditionalDataByReference | Boolean | True | If true, then additional data will be delivered by-reference in the SIP INVITE request. See Section 6 of RFC 7852. |

## 6.6 Device Settings

### 6.6.1 Audio Device Settings

The application shall provide a setting that specifies the audio device to use. The application shall provide a list of audio devices that are available on the computer. The user shall be able to select an audio device from this list.

The default audio device shall be the first audio device in the list of available audio devices.

### 6.6.2 Video Device Settings

The application shall provide the following video device settings.

1. Video device name
2. Image Format (NV12, YUY2, RGB etc.)
3. Image Resolution (320x240, 640x480, etc.)
4. Frame Rate (10, 20, 30 fps)

# 7 Application Logging Requirements

The application software shall write significant events to an application log.

Each log event shall contain the following information.

1. Date/Time (format shall be: 2023-08-26T08:27:53.18042)
2. Application software version
3. Logging level (DEBUG, INFO, WARNING, ERROR, CRITICAL)
4. Class name
5. Class method
6. Message
7. Exception information (may be null)

The application shall use a rolling file appendor. The maximum file size for each application log file shall be 1Mbyte. The maximum number of application log files shall be 5.

The location of the application log files shall be stored underneath the Windows special folder called LocalApplicationData. For instance, if the current user is “John”, the application log files will be located in: C:\Users\John\AppData\Local\OspSimulator\Logs.

# 8 Revision History

## 8.1 Revision 1.0.0 – 6 Apr 2026

| Issue Number | Change Type | Description |
|---|---|---|
| NA | NA | Initial version for software version 1.0.0. |